July 4, 2026

The Editors
[Target journal — e.g. Psychological Methods / Behavior Research Methods]

Dear Editors,

We submit for your consideration the manuscript **"A transdiagnostic dimensional map as a self-calibrating measurement instrument: projection, information, and value of information under structured missingness."** We believe it is well suited to a quantitative-methods venue because its contribution is methodological rather than substantive: it reframes a fitted dimensional model of psychopathology as a *measurement instrument* whose per-patient behaviour — how sharply each person is located, which item would sharpen them next, and how far the reported uncertainty can be trusted — is computable in closed form on a single fitted object.

The core idea. Dimensional maps of psychopathology are typically read as fixed point-coordinates that hide how each patient was measured. We show instead that a missingness-native Bayesian latent-variable model, fit to each patient's observed cells only (no imputation) under mixed likelihoods, returns for every one of $N = 9{,}013$ patients a posterior coordinate *and* its uncertainty, and that four instrument properties follow from that one object by exact identities: (i) a Woodbury-reduced expected-a-posteriori projection that returns the prior — never a false-precise point — on unmeasured axes; (ii) an information accounting showing that what localizes a patient is the Fisher information an item carries, not its loading; (iii) a value-of-information rule, closed-form via the matrix-determinant lemma, for designing efficient batteries; and (iv) a simulation-validated, misspecification-stressed calibration of the reported intervals. A five-corner archetypal simplex then summarizes the continuum as graded membership in named clinical poles.

What is new in this version. We have strengthened the paper specifically on the axis a methods reviewer cares about most — *does the instrument beat the simpler tools a reader already has?* — by adding four head-to-head analyses. (1) Carrying the full posterior coordinate improves out-of-sample prognosis over a naive per-axis sum-score, and the advantage *widens* as measurement grows sparse (the regime where a point estimate is least trustworthy). (2) Ranking battery items by information rather than by loading raises mean reliability by 0.11 at a fixed 27-item budget. (3) The posterior calibration degrades gracefully — never catastrophically — under four controlled violations of the model's own assumptions, converting an internal-consistency check into a robustness result. (4) We report, with equal prominence, two honest null results (a fully patient-adaptive item order coincides with the fixed order; and real-data confidence-triage does not improve one coarse clinical endpoint), because each delimits precisely what the instrument does and does not buy.

We also correct and sharpen the archetype analysis: rather than defend the simplex on reconstruction fidelity — which we now show is the *wrong* yardstick, since the map's near-isotropic variance lets even a random five-dimensional subspace retain more variance than the archetypes — we make the case on what the simplex uniquely provides over principal components: convex, non-negative, per-patient membership in five nameable clinical poles, over a population we confirm has no natural clusters.

Relationship to a companion paper. This manuscript is the methodological half of a two-paper pair. A companion clinical paper (in preparation / under separate submission) reports the substantive findings from the same cohort — the biology-general-factor separation, the durable axis, and the prognostic gradient across archetypes. The two papers share the fitted model but are disjoint in claims: the present paper makes no clinical discovery claim, and the companion makes no methodological one. We are happy to provide the companion manuscript to the editors on request to confirm the absence of overlap.

Declarations. The work uses no external data beyond the cohort described; all analysis code and the fitted-model scoring scripts are released with the paper. The manuscript is not under consideration elsewhere, and all authors approve this submission. We have no competing interests to declare.

We would be glad to suggest reviewers with expertise in item-response and computerized adaptive testing, latent-variable models under missing data, and Bayesian measurement calibration, and we thank you for considering the manuscript.

With best regards,

The Authors